Do Input-Shift Alarms Predict Cross-Survey Performance Loss? Evidence from DHS Child-Mortality Models

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Abstract

A model that predicts child mortality is trained on one demographic and health survey and then reused in a different country or a later year. The standard safeguard against this reuse is out-of-distribution detection: monitor the inputs and raise an alarm when the target population looks different from the training distribution. We ask whether such input-shift alarms actually predict the thing a deployer cares about, namely the loss in performance when the model is transferred. Using 70 Demographic and Health Surveys spanning three decades and a censoring-correct infant-mortality outcome with nationally representative sampling weights, we measure transfer loss as the weighted AUC a local retrain recovers over a transferred model. Transfer loss is real but modest, a mean of 0.010 AUC with 22 percent of transfers losing more than 0.05. Input-shift alarms do not reliably predict this loss: Mahalanobis detectability correlates with transfer loss at r = 0.23, and both Mahalanobis distance and a supervised domain classifier flag high-loss transfers only slightly above chance (AUROC 0.59 and 0.63). The practical conclusion is that input-shift alarms are a weak substitute for measuring performance directly, and deployers of tabular health models should budget for the small amount of local data that monitoring actually requires.

Introduction

Machine learning for global health is trained in one place and deployed in another. A model that estimates the risk that a child dies in infancy is typically fit on a Demographic and Health Survey (DHS) in a single country, or on a pooled set of surveys, and is then expected to score families in a different country or in the same country several years later (Rutstein and Rojas 2006; Croft et al. 2018). This reuse is the default deployment pattern, not an edge case, because ground-truth mortality labels are expensive and building a bespoke model for every target population is often infeasible in resource-constrained settings.

Distribution shift is the central risk in this workflow (Quiñonero-Candela et al. 2009; Koh et al. 2021). The standard safeguard is out-of-distribution detection: monitor the inputs, and raise an alarm when the target looks different from the training distribution (Yang et al. 2021). A large literature has developed detectors for exactly this purpose (Lee et al. 2018; Liu et al. 2020; Sun et al. 2022; Liu, Ting, and Zhou 2008). This paper asks a narrower and more practical question than whether these detectors can see that the target differs: **does an input-shift alarm tell a deployer when a transferred model will actually lose performance?**

Detecting that two populations differ is not the same as predicting that a model will fail. The two tasks are distinct, and prior work has noted the gap (Quiñonero-Candela et al. 2009). Our contribution is to measure this gap directly, at scale, for a real deployment setting. We make three points. First, we build a censoring-correct child-mortality outcome, infant death before twelve months, so that every included child is either observed to die in infancy or observed to survive it. Second, we measure transfer loss across 70 surveys and 552 source-to-target pairs using nationally representative sampling weights, and find that it is real but modest. Third, we show that input-shift alarms, both unsupervised Mahalanobis distance and a supervised domain classifier, predict this loss only weakly, flagging high-loss transfers slightly above chance. The implication is that input-shift detection is a weak substitute for measuring performance directly, and that a deployer is better served by a small budget of local labels than by a sophisticated alarm.

Data and Outcome

We use the DHS Individual Recode files, the standard women's survey instrument collected across low- and middle-income countries (Rutstein and Rojas 2006; Croft et al. 2018). The Individual Recode records one row per interviewed woman and embeds the full birth history as wide arrays, one set of columns per birth. We reshape these arrays into child-level records and retain the 70 surveys with at least 30 observed infant deaths, spanning 68 countries and DHS phases II through IX, roughly 1990 to 2022. Two further surveys are excluded because their files omit the age-at-death variable.

The outcome definition is central to the validity of the analysis. A naïve label that takes a child alive at interview as a survivor to age five is wrong, because a child alive at six months has not been observed for the full five years, and differing follow-up lengths across surveys can manufacture the appearance of drift. We therefore use a fixed, fully observed horizon. We restrict to children born 12 to 59 months before the interview, so that every child has been observed for the

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full first year of life, and we label a death as infant death if the age at death is below 12 completed months. Every included child is then either observed to die in infancy or observed to survive it, which removes the censoring artifact.

Because the DHS uses a complex survey design, we weight every estimate. Each child inherits the mother's sample weight, the standard approach for child-level DHS analyses (Croft et al. 2018). We pass these weights to the training loss and evaluate with a weighted Mann-Whitney AUC and a weighted Brier score, so that all reported rates and distributions are nationally representative.

Methods

Transfer Loss

For each survey we train a gradient-boosted tree model (Chen and Guestrin 2016) to predict infant death from six features available in every DHS phase: child sex, birth order, preceding birth interval, mother's age at birth, and maternal education in level and in years. We measure ranking quality with weighted area under the ROC curve (AUC). In-domain performance is the weighted three-fold cross-validation AUC on a survey. For a source-to-target pair, transfer performance is the weighted AUC of the source model applied to the target, and *transfer loss* is the difference between the target's in-domain AUC and the transferred AUC. Transfer loss is nonnegative in expectation and measures how much ranking performance a deployer leaves on the table by not retraining locally.

Input-Shift Alarms

We evaluate two kinds of input-shift alarm. The first is unsupervised Mahalanobis distance (Lee et al. 2018), computed from a Gaussian fit to the source features. The second is a supervised domain classifier, a gradient-boosted model trained to distinguish source from target on the features alone (Quiñonero-Candela et al. 2009). For each source-to-target pair we score the detectability of the target as the AUC of each alarm in separating source from target, and we ask whether that detectability predicts the transfer loss.

We select eight source surveys to span geography and survey phase: Kenya, Nigeria, Bolivia, Egypt, Indonesia, the Philippines, Rwanda, and Haiti. For each source we pair it with all 69 remaining surveys, giving 552 source-to-target pairs. This is the full set of transfer directions for these sources, not a subsample, and we report it as such.

Results

Transfer Loss Is Real but Modest

Weighted in-domain AUC averages 0.566 across the 70 surveys, and transferred AUC averages 0.556. Transfer loss has mean 0.010 and median 0.018, with 64 percent of pairs losing performance and 22 percent losing more than 0.05 AUC. Figure 1 shows the distribution. The loss is small in absolute terms because infant death is only weakly predictable from tabular covariates, which is itself an important fact: the signal is fragile, but so is the amount that can be lost.

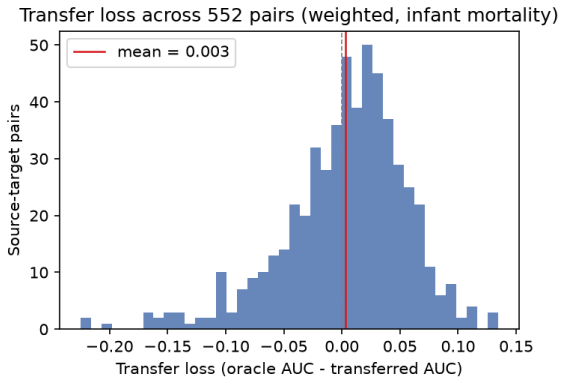


Figure 1: Distribution of transfer loss across 552 source-to-target pairs. Transfer loss is the weighted AUC a local retrain recovers over a transferred model. The mean is 0.010 and 22 percent of pairs lose more than 0.05 AUC.

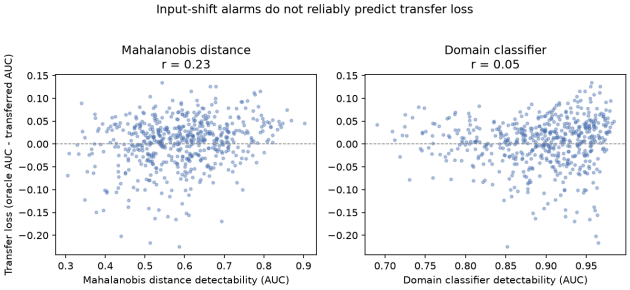


Figure 2: Input-shift detectability does not reliably predict transfer loss. Each panel plots, for 552 source-to-target pairs, an alarm's out-of-distribution score (x-axis) against transfer loss (y-axis). Correlations are r = 0.23 for Mahalanobis distance and r = 0.05 for a domain classifier.

Input-Shift Alarms Predict Loss Only Weakly

Figure 2 plots each alarm's detectability against transfer loss. Mahalanobis detectability correlates with loss at r = 0.23, and the domain classifier at r = 0.05. The relationship is weak in both cases, and the supervised domain classifier, which is the better shift detector, is the weaker loss predictor because it saturates near perfect separation for most country pairs. The overall correlation is an average over eight sources whose individual correlations range from r = -0.09 to $+0.47$ for Mahalanobis and from -0.25 to $+0.35$ for the domain classifier, so no single source drives the result, and for several sources the relationship is even negative.

The more operationally meaningful question is whether an alarm can flag the transfers that matter, namely those with a prespecified meaningful loss. Figure 3 reports the AUROC of each alarm for flagging transfers with loss above 0.05. Mahalanobis distance reaches 0.59 and the domain classifier 0.63, only slightly above the 0.50 chance baseline. Raising an alarm on the top fifth of Mahalanobis scores flags high-loss transfers at a precision of 0.19, against a baseline prevalence of 0.16. In short, an input-shift alarm is a weak substitute for

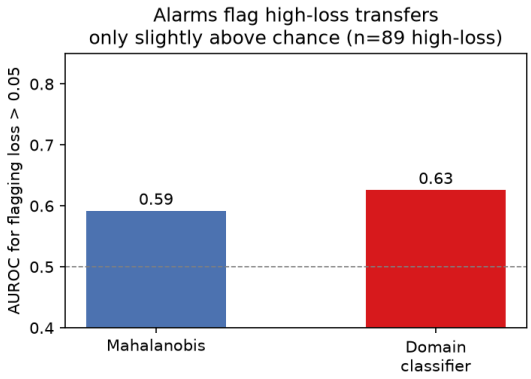


Figure 3: AUROC of each input-shift alarm for flagging source-to-target pairs whose transfer loss exceeds 0.05 AUC. Both alarms are only slightly above chance, so they are unreliable safety mechanisms.

Alarm	r vs. loss	AUROC	r range
Mahalanobis	0.23	0.59	$[-0.09, 0.47]$
Domain clf.	0.05	0.63	$[-0.25, 0.35]$

Table 1: How well input-shift alarms predict transfer loss across 552 source-to-target pairs. AUROC is the alarm's ability to flag transfers with loss above 0.05, and the r range is across the eight sources. Both alarms are weak, and several per-source correlations are negative.

measuring performance directly.

Related Work

Out-of-distribution detection has concentrated on vision and language, where the standard recipe fits a model on one dataset and scores the distance of a test input to the training distribution (Lee et al. 2018; Liu et al. 2020; Sun et al. 2022), with unsupervised anomaly detectors such as isolation forest as a baseline (Liu, Ting, and Zhou 2008). Distribution shift decomposes into covariate shift, where the inputs change, label shift, where the class prevalence changes, and concept drift, where the conditional outcome changes (Quiñonero-Candela et al. 2009; Gama et al. 2014). Prior work notes that detecting shift is distinct from predicting its harm (Quiñonero-Candela et al. 2009). Our contribution is to quantify that distinction for tabular health data, where the inputs change little while the outcome relationship changes, the regime in which input-based detectors are least able to help.

Machine learning for child mortality has mostly treated prediction as a one-time exercise (Bountogo et al. 2023). We study the deployment step itself, and we do so with an outcome definition that matches the DHS life-table convention of restricting to fully exposed children (Croft et al. 2018).

Limitations

The transfer loss is measured with gradient-boosted trees; other estimators may shift the absolute numbers, though the weak alarm-to-loss relationship should persist. The eight sources are chosen for diversity but are not all surveys, and the conclusion should be read as covering these directions. The analysis applies sampling weights but does not model within-cluster correlation. Finally, our alarm evaluation uses a retrospective sample of targets; a prospective deployment would raise alarms online, which is the natural next step.

Conclusion

We asked whether an input-shift alarm tells a deployer when a transferred child-mortality model will lose performance. The answer, on a censoring-correct infant-mortality outcome across 70 surveys and 552 pairs, is largely no: transfer loss is real but modest, and input-shift alarms predict it only weakly, flagging high-loss transfers slightly above chance. For deployable AI, this means that monitoring the inputs is a weak substitute for monitoring performance, and that a small budget of verified local labels, whose exact size we leave to future work, is likely the more reliable safeguard.

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